

High-Dimensional Global Optimization Method for High-Frequency Electronic Design

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Abstract—Efficient global optimization of microwave systems is a very challenging task that emerges in importance for rapid design closure and discovery of novel structures. As the operating frequency increases, additional difficulties in design optimization occur due to increased nonlinearity, creating a high-dimensional nonconvex response surface. Bayesian optimization (BO) is a promising solution to solve such problems. However, BO-based methods suffer from the curse of dimensionality, where the number of simulations required for convergence increases exponentially with the number of parameters. In this paper, we address this problem and propose a new BO-based high-dimensional global optimization method titled, Bayesian Optimization with Deep Partitioning Tree (DPT-BO). DPT-BO leverages a novel DPT that allows for rapid coverage of high-dimensional sample spaces and utilizes an additive Gaussian process (ADD-GP) with a fully additive decomposition, making it more suitable for high-frequency design optimization. We apply DPT-BO to different optimization test functions along with three high-frequency design applications, namely, maximizing signal integrity in high-speed channels, minimizing losses of substrate integrated waveguides with air cavity, and maximizing efficiency of wireless power transfer systems. The results show that DPT-BO finds control parameters that provide better performance in less CPU time compared to other techniques.

Index Terms—Air-filled substrate integrated waveguide (SIW), Bayesian optimization (BO), high-dimensional global optimization, millimeter wave, signal integrity, wireless power transfer (WPT).

I. INTRODUCTION

MINIATURIZATION of systems has created a trend toward increasing the operating frequency of such systems with increased bandwidth but in a smaller area. As the operating frequency increases; however, such systems tend to become more sensitive to variations and interactions between their control parameters, such as geometrical dimensions and material properties. This raises new challenges related to design optimization, especially when the number of such parameters increases due to interactions and nonlinearity. Approximations made using analytical models often tend

to provide erroneous results and unable to capture these interactions. Accurate models used in optimization problems therefore need to rely on accurate electromagnetic (EM) simulations. Since such simulations are computationally expensive, the problem is further exacerbated. From the optimization perspective, the solution therefore corresponds to minimizing or maximizing a nonconvex, high-dimensional response surface where the function queries are generated from an EM simulator. Since the analytical form of the response surface is not available, the optimization needs to be conducted in the black-box setting.

Often times, conventional techniques are not suitable for optimization of such systems due to many difficulties. Gradient-based methods, for instance, require approximating the gradient in a black-box setting, which can degrade the convergence rate and can quickly become computationally intractable. When the problem is nonconvex, as common with high-frequency electronics, the algorithm needs to be initiated with random restarts to avoid getting stuck in many local optima points. Evolutionary algorithms, such as genetic algorithm and differential evolution, are commonly used alternatives. However, such techniques require a large number of generations and mutations to ensure convergence to the global optimum.

In recent years, surrogate-based optimization methods have gained attention in high-frequency design optimization problems to address some of the aforementioned challenges. Here, a finite amount of training data is collected using CPU intensive simulations, and a predictive model is trained to replace the EM solver. Popular modeling techniques in this area include artificial neural networks (ANN) [1], [2], support vector machines (SVM) [3], polynomial chaos (PC) [4], and Gaussian Processes (GP) [5], [6]. Once the model is derived, it can be combined with optimization techniques since the surrogate or meta models can be used to generate vast amounts of data. However, the generation of these models often times requires large amounts of training data, especially when the dimensionality increases, thereby limiting their use in high-dimensional optimization problems. Knowledge-based methods utilizing approximate analytical expressions [7] and/or equivalent circuits [8] are commonly used to reduce the amount of training data required to maintain high predictive accuracy. Among them, space mapping (SM) [9], a prediction-correction based local optimization method, is widely recognized and is one of the more popular techniques in microwave engineering with numerous successful results reported in the literature [10]–[12]. However, convergence and final values

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obtained using SM-based approaches heavily rely on the correlation between the coarse and fine model, as well as the choice of the transformation, which often times is linear. We refer readers to [13] for a recent overview of SM-based approaches and a more detailed discussion on SM-related challenges.

A promising method with strong theoretical foundations to overcome these challenges is Bayesian optimization (BO)-based on GPs [14]. BO is a well-studied method in various disciplines and has attracted recent attention in optimization of electronic systems [15]–[17]. Here, the goal is to directly find the global optimum rather than deriving a high-quality predictive model covering the entire sample space, thus BO-based methods require less CPU intensive simulations compared to deriving a predictive model first and then optimizing it. However, efficient high-dimensional BO (HDBO) is still an open problem in the literature.

In this paper, we propose a new method to scale BO to high-dimensional problems, namely, Bayesian Optimization with Deep Partitioning Tree (DPT-BO). In this method, the objective is to reduce the computational complexity through a reduction in the number of EM simulations required for the optimization. Motivated by recent advances in HDBO [18]–[20], we adopt the use of additive GP (ADD-GP) and approximate the objective function as a summation over lower dimensional group of functions, each depending on a subset of input parameters [21]. Previous methods utilizing ADD-GP assume that the function decomposes into nonoverlapping groups, meaning that the parameters in a particular group can be optimized independently without considering the parameters in other groups. This is mainly to handle computational complexity associated with the auxiliary optimization of the nonconvex acquisition function used in the BO framework. However, nonoverlapping groups make use of an assumption that is invalid in high-frequency design in that high-dimensional systems tend to have interacting variables. For instance, geometrical parameters of an antenna and its matching network cannot be optimized independently of each other. Hence, we propose using a fully additive decomposition where all interactions at a particular order are preserved, allowing to capture much richer classes of functions. To the best of authors' knowledge, the fully additive structure that allows for capturing interactions between every parameter has not been reported in the context of BO in the open literature.

To address the challenge of auxiliary optimization of the acquisition function, we extend the hierarchical partitioning tree approach in [22] and propose a novel *deep hierarchical partitioning tree*. The proposed strategy eliminates the auxiliary optimization step from the BO framework and allows for a rapid coverage of the high-dimensional sample space. We show the performance DPT-BO on several optimization test functions consisting of many local optima and different response surfaces, as well as three applications arising in high-frequency electronic design, namely: 1) maximizing signal integrity in high-speed channels; 2) minimizing losses over D-band for substrate integrated waveguides (SIW) with an air cavity; and 3) maximizing efficiency of a wireless power transfer (WPT) system. The MATLAB implementation

of DPT-BO used in these applications is publicly available at <https://github.com/hakkimerttorun/DPTBO>.

The rest of the paper is structured as follows. Section II provides the background and formulation for BO based on GPs. Section III presents the new DPT-BO algorithm with application to test functions. Section IV demonstrates the application of DPT-BO for optimizing interconnects on a silicon interposer high-speed channel. Section V presents the minimization of transmission losses of an SIW structure with air cavity operating in the D-band. Section VI addresses the multiobjective optimization of WPT system followed by the conclusion in Section VII.

II. GAUSSIAN PROCESSES AND BAYESIAN OPTIMIZATION

BO is a class of active learning-based machine learning (ML) algorithms. Here, the next sampling point is selected to maximize the reward toward finding the global optima of a nonconvex, black-box function, given as

$$\tilde{x} = \arg \max_{x \in \mathbb{X}^D} f(x) \quad \text{or} \quad \tilde{x} = \arg \min_{x \in \mathbb{X}^D} f(x) \quad (1)$$

where $f(x)$ is the objective function, x is the input vector consisting of control parameters of the design to be optimized, $\tilde{x}$ is the global optimizer, and $\mathbb{X}^D$ is the D dimensional bounded sample space, i.e., $x \in \mathbb{X}^D \subset \mathbb{R}^D$. The approach is based on Bayes' Theorem

$$P(y|D_{1:t}, \theta) = \frac{P(D_{1:t}|y, \theta)P(y|\theta)}{P(D_{1:t}|\theta)} \quad (2)$$

where $y_i = f(x_i) + \epsilon_i$ are noisy observations of the objective function in (1); $D_{1:t} = [(x_1, y_1), (x_2, y_2), \dots, (x_t, y_t)]$ represent the data set obtained after t th iteration; θ is the collection of model parameters, i.e., hyperparameters; $P(y|\theta)$, $P(y|D_{1:t}, \theta)$, and $P(D_{1:t}|y, \theta)$ are distributions for the prior, posterior, and likelihood conditioned on the model parameters, respectively. For the purposes of design optimization, observations will generally be made through deterministic EM and/or circuit simulations. We consider the noisy setting and set $\epsilon \sim \mathcal{N}(0, \sigma_n^2)$ to account for possible numerical errors in the simulation framework, e.g., small meshing errors involved in full-wave EM simulations utilizing adaptive meshing techniques.

There are several possible choices of probabilistic models that can be used as the prior in (2). Some of these include Bayesian neural networks (BNN) [23], random forest regression [24], and parzen estimators [25]. However, these methods have relatively poor capability of predicting the posterior in (1), especially in the scarce data regime that is sought after in high-frequency design optimization problems [26]. GPs, on the other hand, have many theoretical and practical advantages that align with the purposes of BO. From the theoretical point of view, it has been shown that BNN with one hidden layer converges to a GP when the number of hidden units tends to infinity [27]. Compared to BNNs, GPs are nonparametric models and have much smaller number of trainable hyperparameters. The nonparametric Bayesian nature of GPs also automatically implements Occam's Razor [28] and limits the complexity of the model. When this is combined

with a small number of learnable hyperparameters, GPs tend to avoid overfitting to scarce data, which makes them a very promising choice for the purposes of BO. Furthermore, GPs with a certain type of kernels have the universality property, meaning that they can model any function over a compact set when the number of training data tends to infinity [29], [30].

Formally, GPs are distributions over functions that model underlying function to be optimized, $f(x)$, as a collection of infinitely many random variables, any finite number of which having a joint Gaussian distribution [31], given by

$$f(x) \sim \mathcal{N}(\mu(x_{1:t}), K_{1:t}) \quad (3)$$

where $\mu(x)$ is the mean function and K is the covariance matrix, constructed by a kernel function $k(x, x')$. The mean function introduces a strong parametric bias to, otherwise, fully nonparametric setting. In many problems, this limits the generalization capability of the GP model since predictions far away from the training data will converge to the prespecified parametric mean function [32]. Although this can be useful for some applications, for the purposes of BO, it is common to use a constant mean function [33], i.e., $\mu(x) = m$.

Since the mean function is set to a constant, the kernel function solely determines types of interactions that the final model can capture. Domain knowledge of the underlying function can be captured by choosing a descriptive kernel function. For instance, the popular squared exponential (SE) kernel is used for infinitely differentiable smooth functions, whereas Matern-type kernels are used when the function is believed to be less smooth. A commonly used kernel for BO is Matern 5/2 function with automatic relevance determination (ARD) [33], given as

$$k(x_i, x_j) = \sigma_f^2 \left(1 + \sqrt{5}r + \frac{5}{3}r^2 \right) e^{-\sqrt{5}r} \quad (4)$$

where

$$r = \left(\sum_{d=1}^D \frac{(x_{i,d} - x_{j,d})^2}{\lambda_d^2} \right)^{1/2}$$

λ_d is the lengthscale of each input parameter and σ_f is the signal standard deviation. The value of constant mean function (m), noise standard deviation (σ_n), λ_d , and σ_f are collectively referred to as θ , hyperparameters of the GP model.

After defining the mean and kernel function, the training of the GP is performed to find the hyperparameter vector. This is done by using a quasi-newton-based gradient descent method to minimize the negative log marginal likelihood function, given by:

$$\log p(y_{1:t} | x_{1:t}) = -\frac{1}{2}(y_{1:t} - m)^T (K + \sigma_n^2 I)^{-1} (y_{1:t} - m) - \frac{1}{2} \log |K + \sigma_n^2 I| - \frac{t}{2} \log 2\pi \quad (5)$$

where I is the identity matrix of size t . Once the training is done, the predictive posterior distribution at a test point x^* can be written as

$$p(y^* | D_{1:t}, x^*, \theta) = \mathcal{N}(y^* | \hat{\mu}(x^*), \hat{\sigma}^2(x^*)) \quad (6)$$

$$\hat{\mu}(x^*) = m + k^T (K + \sigma_n^2 I)^{-1} (y_{1:t} - m) \quad (7)$$

$$\hat{\sigma}^2(x^*) = k^* - k^T (K + \sigma_n^2 I)^{-1} k \quad (8)$$

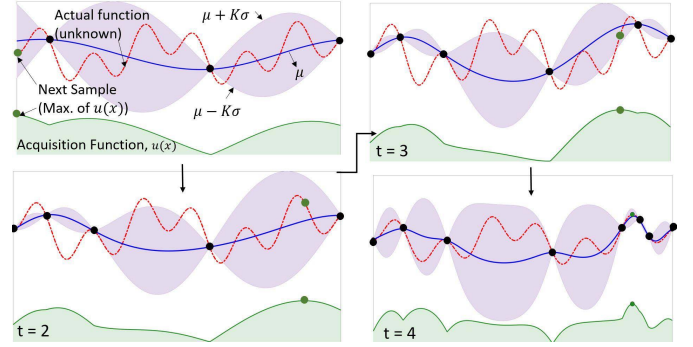


Fig. 1. Illustration of BO on maximizing a toy function [34].

where $k^* = k(\tilde{x}^*, x^*)$ and $k = k(x^*, x_{1:t})$ are as in (4). In the BO framework, the pointwise predictive mean and variance in (7) and (8) are used to construct a predefined sampling strategy, referred as the acquisition function, $u(x)$. Most popular functions are probability of improvement (PI), expected improvement (EI), and upper confidence bound (UCB), given as

$$u_{PI} = \Phi\left(\frac{(\mu(x) - \tilde{f}^* - \zeta)/\sigma(x)}{\sigma(x)}\right) \quad (9)$$

$$u_{EI} = (\mu(x) - \tilde{f}^* - \zeta) \Phi(Z) + \sigma(x) \phi(Z) \quad (10)$$

$$u_{UCB} = \mu(x) + K\sigma(x), \quad K = \sqrt{2\ln(2\pi M^2/(12\eta))} \quad (11)$$

where $\tilde{f}^*$ is the best point observed so far, ζ is a hyperparameter of u_{PI} and u_{EI} , M is number of calls made to UCB so far, $(1 - \eta)$ is the probability of zero regret for UCB [35], $Z = (\mu(x) - \tilde{f}^* - \zeta)/\sigma(x)$, $\Phi(\cdot)$ and $\phi(\cdot)$ are the CDF and PDF of normal distribution, respectively. After $u(x)$ is defined, the next set of control parameters to be simulated, x_{t+1} , is selected as the input vector maximizing $u(x)$ as a result of an auxiliary optimization procedure. Note that although $u(x)$ is nonconvex, it is fast to evaluate, hence, conventional methods such as DIRECT [36] can be used. The iterative procedure of BO for maximizing a toy function is given in Fig. 1 and a more detailed background can be found in [14].

III. HDBO WITH DEEP PARTITIONING TREE

There are two main challenges in efficiently scaling BO to higher dimensional problems: 1) statistical challenge of approximating the objective function, $f(x)$ and 2) auxiliary optimization of the nonconvex acquisition function. The former is related to the need for exponentially more observations to establish a certain accuracy of the nonparametric GP model, i.e., curse of dimensionality. The latter is a computational challenge since this auxiliary optimization itself is exponentially difficult in higher dimensions. Addressing these two challenges efficiently requires making assumptions on the structure of $f(x)$ to efficiently handle the optimization complexity.

In this section, we first present the structural form of $f(x)$ we use in DPT-BO and compare with the existing forms in the literature. Then, we present the DPT as a sampling strategy to select the next system simulation parameters and address the auxiliary optimization challenge.

A. Model Structure

A substantial portion of the prior work in HDBO assumes the objective function lies in a lower dimensional subspace of the original sample space. Here, attempts are mainly based on approximating the original function with a lower dimensional embedding [37], [38]. However, this is a very strict assumption for high-frequency design optimization since the high dimensionality can arise due to a combination of device- and circuit-level components' parameters such as transistor size, lumped element parameters, and geometrical parameters of EM structures, all of which significantly affect the system performance.

Recently, this assumption was relaxed by assuming that $f(x)$ admits an additive form, thus allowing it to depend on all input variables. Functions of this form can be written as [18]

$$f(x) = f^{(1)}(x^{(1)}) + f^{(2)}(x^{(2)}) + \dots + f^{(M)}(x^{(M)}) \quad (12)$$

where $x^{(j)}$ represent parameters that are decomposed into M disjoint groups such that $x = \bigcup_{j=1}^M x^{(j)}$ and $x^{(j)} \cap x^{(i)} = \emptyset$. This structure can be realized at the kernel level of the GP by parameterizing the kernel function as

$$k(x, x') = k^{(1)}(x^{(1)}, x'^{(1)}) + k^{(2)}(x^{(2)}, x'^{(2)}) \dots \dots + k^{(M)}(x^{(M)}, x'^{(M)}) \quad (13)$$

where $k(x, x')$ is the D dimensional kernel used to construct the covariance matrix in (3). Note that the structure in (12) simplifies the auxiliary optimization of the acquisition function. Since $u(x)$ admits the same structure, each group of $u^{(j)}(x^{(j)})$ can be optimized separately. Several works are built on this formulation to efficiently learn the decomposition [19] or relax the disjoint assumption by allowing a limited degree of interaction between groups [20]. However, the underlying structural assumption in (12) is still conservative since grouping the input variables in such a way means that $f(x)$ can be optimized by finding the optimal parameters in each group independently of the parameters in other groups, which is hardly the case in high-frequency designs.

In this paper, we further relax the structural assumption and propose using a fully additive decomposition to preserve the interactions between variables. Here, the additive structure we consider corresponds to high-dimensional model representation (HDMR) [39] and can be written as

$$f(x) = \sum_{i=1}^D f_i(x_i) + \sum_{1 < i < j < D} f_{ij}(x_i, x_j) + \sum_{\substack{1 < i_1 \\ \dots < i_n < D}} f(x_{i_1}, \dots, x_{i_n}) \quad (14)$$

where n denotes the maximum allowed order of interactions among D possible orders. Since the GP we use has a constant mean function, the structure in (14) can be realized via the

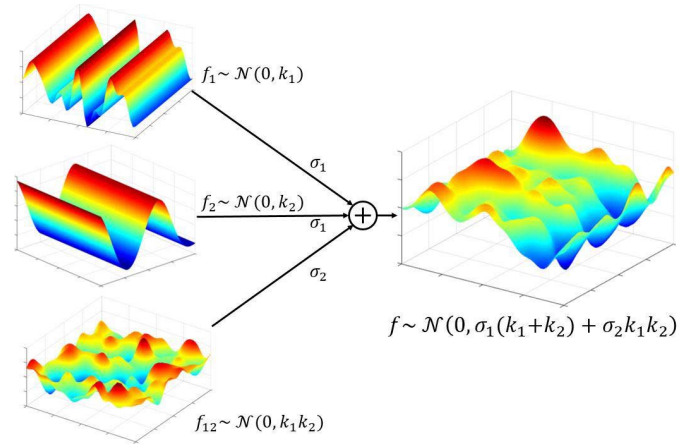


Fig. 2. 2-D illustration of the additive decomposition used by DPT-BO.

fully additive kernel of ADD-GP, written as [21]

$$k(x, x') = \sigma_{f_1}^2 \sum_{i=1}^D k_i(x_i, x'_i) + \sigma_{f_2}^2 \sum_{1 < i < j < D} k_i(x_i, x'_i) k_j(x_j, x'_j) \dots + \sigma_{f_n}^2 \sum_{\substack{1 < i_1 \\ \dots < i_n < D}} \left[\prod_{d=1}^n k_{i_d}(x_{i_d}, x'_{i_d}) \right] \quad (15)$$

where $k_i(x_i, x'_i)$ is the subkernel of each parameter, chosen as Matern 5/2 as in (4) with unit variance; $\sigma_{f_n}^2$ is the signal variance hyperparameter associated with each order of interaction, enabling to learn the contribution of each level of interaction to the overall predictive GP. It should also be noted that as each input variable has a separate kernel, a separate lengthscale is assigned to every parameter, effectively implementing ARD. The prior associated with the kernel function in (15) is illustrated in Fig. 2 for the 2-D case.

The number of terms in the right-hand side of (14) increases superexponentially with D according to the *Bell Number*, which is 115975 for $D = 10$. Hence, we only preserve up to R th order of interactions in (15) along with the D th order of interaction corresponding to a standard GP model, i.e., $n = 1, 2, \dots, R, D$. This makes the proposed DPT-BO algorithm a superset of the BO methods using a standard GP. Here, during the training of the GP model, the likelihood in (5) can favor increasing σ_D^2 and decreasing $\sigma_{1, \dots, R}^2$ for a particular problem and vice versa for others. As the proposed DPT-BO algorithm updates the hyperparameters at every iteration, this corresponds to dynamically weakening or strengthening the structural assumption in (14) as the optimization progresses and allows for considering much richer classes of functions compared to prior work in the literature.

B. Deep Hierarchical Partitioning Tree

A hierarchical partitioning tree is a sampling strategy that is utilized in nonconvex global optimization algorithms [40]. Here, the original sample space, $\mathcal{X}^D$, is divided into several child nodes, each containing a disjoint subspace of $\mathcal{X}^D$. The objective function, $f(x)$, is then queried at each node to select the next branch to be expanded. In the context of BO, several works have considered such a strategy to eliminate

the auxiliary optimization step from the BO framework [22], [41], [42]. Particularly, the strategy in [22] only uses the acquisition function to consider each subspace as a candidate region that possibly contains the global optima. This allows for determining the most promising region without any function queries. Although this provides a fast coverage of a sample space with moderate dimensionality, the approach in [22] has limited capability of covering a high-dimensional sample space since it requires exponentially more samples to identify a small enough region to be exploited.

In this paper, we build on the candidate region generation strategy of [22] with a novel DPT. Unlike conventional partitioning structures where the child nodes are only expanded in the vertical direction, the proposed strategy also performs expansion in the horizontal direction, thus creating a deep structure.

Here, at every iteration, we group the input parameters according to their sensitivity of creating a variation on $f(x)$. As the kernel we use in (15) has the ARD property, the input variables having a lower lengthscale value have higher impact on $f(x)$ based on the correlation matrix in (3). This provides a cost-free impact analysis, which can be used to perform the grouping as

$$\left\{ x^D = \bigcup_{j=1}^M x_t^{(j)} \mid |x_t^{(j)}| = d \ll D \right\} \quad (16)$$

and

$$\left\{ \lambda^D = \bigcup_{j=1}^M \lambda_t^{(j)} \mid \max \lambda_t^{(i)} < \min \lambda_t^{(j)}, \forall i < j < M \right\} \quad (17)$$

where x^D is the D dimensional input vector; $x_t^{(j)}$ is the j th group of input variables at t th iteration; M is the total number of groups; d is the maximum number of parameters in a particular group, and $\lambda_t^{(j)}$ represents the lengthscale vector of the input parameters in the j th group. Note that as the training of the GP is done at every iteration and the hyperparameter vector is updated, input variables in each group changes dynamically according to (17).

Once the input parameters are grouped, the sample space X^D is divided into 2^d regions of hypercubes, $H_{t,k}$, along each parameter in a particular group, $x_t^{(j)}$. We refer to this division as the *vertical expansion* of the partitioning tree, which is repeated for each group, resulting in a total of $M2^d$ new regions at every iteration. Then, candidate points are generated as

$$c_{t,k} = \frac{H_{t,k,\min} + H_{t,k,\max}}{2}, \quad k = 1, 2, \dots, tM2^d \quad (18)$$

where $c_{t,k}$ is the k th candidate point of a total of $tM2^d$ points after iteration t ; $H_{t,k}$ is the region that $c_{t,k}$ belongs to, and $H_{t,k,\min}$ and $H_{t,k,\max}$ are the corresponding lower and upper boundaries of each D dimensional region.

After the vertical expansion, a possible sampling strategy can be to select the next sampling point as the candidate point maximizing the acquisition function. However, the candidate points generated after the vertical expansion belong to a relatively larger region. Such an approach would provide a

Algorithm 1 Horizontal Expansion in DPT-BO

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1: (Initialize)    $H_t^{(1)} \leftarrow \tilde{H}_t, i \leftarrow 1.$ 
2: (Update  $H_t^{(i)}$ )  $H_t^{(i)} \leftarrow \bigcup_{j=1}^{2^d} h_{t,j}^{(i)}$  along  $\forall x \in x_t^{(i)}$ 
3: (Update  $c_{t,j}^{(i)}$ )  $c_{t,j}^{(i)} \leftarrow (h_{t,j,\min}^{(i)} + h_{t,j,\max}^{(i)})/2,$ 
4: (Select  $c_{t,j^*}^{(i)}$ )  $c_{t,j^*}^{(i)} \leftarrow \arg \max_{c \in C} u(c)$ 
5: (Set  $H_t^{(i+1)}$ )  $H_t^{(i+1)} \leftarrow h_{t,j^*}^{(i)}$ 
6: if  $i = M$  then
7:    $x_{t+1} \leftarrow c_{t,j^*}^{(M)}$ 
8: else
9:    $i \leftarrow i + 1$  and go to Step 2.
10: end if
  
```

fast exploration of the sample space since it identifies the most promising region among many other large regions that comprise the high-dimensional sample space. Yet, it would lack exploitation property since a subsequent search within this large promising region is not performed. This would limit the convergence rate in high-dimensional problems since more simulations would be required to explore a significant portion of the sample space such that the identified promising region becomes small enough.

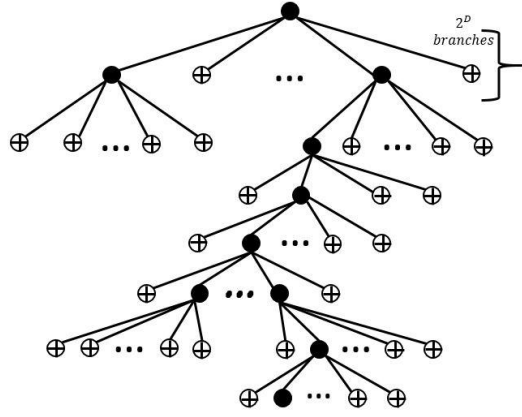
To address the exploitation problem without giving up the exploration capability, we propose a horizontal expansion strategy following the vertical expansion to search within these relatively large regions. First, we select the most promising region that is generated after the vertical expansion as

$$c_{t,j^*} = \arg \max_{c \in C} u(c), \quad c_{t,j^*} \in \tilde{H}_t \quad (19)$$

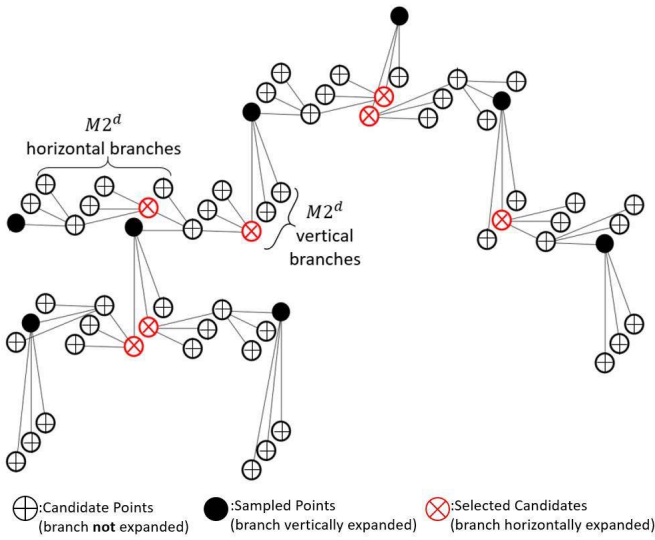
where C is the set of previously generated candidate points; $u(c)$ is the value of the acquisition function, and $\tilde{H}_t$ is the relatively large region that c_{t,j^*} belongs to. We then iteratively shrink $\tilde{H}_t$ by dividing it into 2^d regions along the parameters in a particular group, $x_t^{(i)}$, and generate the candidate points within these regions as

$$H_t^{(i)} = \bigcup_{j=1}^{2^d} h_{t,j}^{(i)}, \quad c_{t,j}^{(i)} = \frac{h_{t,j,\min}^{(i)} + h_{t,j,\max}^{(i)}}{2} \quad (20)$$

where $h_{t,j}^{(i)}$ and $c_{t,j}^{(i)}$ denote the resulting regions and corresponding candidate points when $H_t^{(i)}$ is divided into 2^d new regions. This is followed by selecting the most promising region, $h_{t,j^*}^{(i)}$, from these newly generated ones as in (19) and further dividing it into 2^d new regions along the parameters in the next group, $x_t^{(i+1)}$, as in (20). The horizontal search within $\tilde{H}_t$ is performed M times starting from the first group, i.e., $i = 1$. This corresponds to a sensitivity aware search that prioritizes the input parameters having more impact on $f(x)$ since $x_t^{(i)}$ have more impact on $f(x)$ than $x_t^{(i+1)}$ according to (17) and $x_t^{(i)}$ determine the region to be further divided along $x_t^{(i+1)}$. Note that, at t th iteration, the total number of candidate points generated by DPT-BO corresponds to $2tM2^d$ after the vertical expansion followed by the horizontal one.



(a)



(b)

Fig. 3. Comparison of partitioning tree approaches. (a) Tree structure with only vertical expansion [22]. (b) Proposed Deep Partitioning Tree (DPT).

Finally, the next sampling point is selected as the candidate point that provides the largest acquisition function value after M th division as

$$x_{t+1} = \arg \max_{c \in h_{t,j}^{(M)}} u(c). \quad (21)$$

This horizontal expansion procedure can be implemented using a recursive formulation as given in Algorithm 1. An example partitioning tree constructed by DPT-BO is given and compared to conventional vertical only strategy in Fig. 3.

After performing the simulation at x_{t+1} , the region that x_{t+1} belongs to, $h_{t,j^*}^{(M)}$, is vertically expanded to proceed into next iteration as

$$H_n = \bigcup_{i=1}^{M2^d} h_i, \quad h_i \subset h_{(t,j^*)}^{(M)} \quad (22)$$

$$H_{t+1} = H_n \cup H_t \quad (23)$$

where H_n is the union of new regions, h_i , acquired by dividing $h_{t,j^*}^{(M)}$ into $M2^d$ hypercubes.

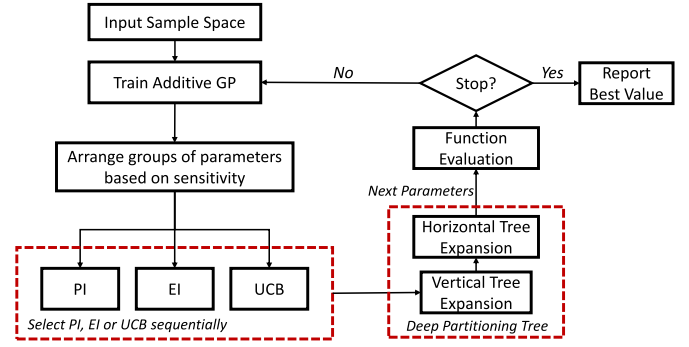


Fig. 4. Overall flowchart of the proposed DPT-BO method.

C. Selecting Acquisition Function

Determining which acquisition function to use is a vital part of any BO-based algorithm. In recent years, information-theoretic entropy-based acquisition functions, such as predictive entropy search (PES) [43], have gained particular interest in BO community. Such methodologies aim to directly estimate $\arg \max$ of the function and is shown to be particularly useful as compared to improvement-based PI, EI, and UCB for problems with moderate dimensionality. However, for high-dimensional problems, information-theoretic approaches become computationally intractable since such methodologies involve estimating intractable distributions. This requires either making poor approximations, such as mean-field approximation, which can result in poor convergence or resorting to sampling-based techniques that introduce additional computational complexity.

In this paper, we therefore use improved-based acquisition functions. However, there is no guarantee that a single improvement-based strategy will outperform others at every iteration and for every problem. For instance, PI in (9) is biased toward exploitation over exploration. This limits its use for nonconvex problems with many local optima, however, very advantageous when the objective function has a narrow valley, such as encountered in microwave filters [44]. On the other hand, EI and UCB in (10) and (11) can better handle the exploration and exploitation tradeoff in larger sample spaces compared to PI, e.g., material and geometry cooptimization problems with a large selection of possible dielectrics that can be used.

To address this issue, a probabilistic gain-based strategy that uses a portfolio of acquisition functions was proposed [45], which was extended in [22] to a deterministic observation-based strategy. Both strategies rely on a reward function to determine which acquisition function performs the best for the given problem. However, a reward-based strategy can be misleading in high-dimensional problems since the reward function can be biased toward a particular $u(x)$ in the early iterations of the optimization. Since this bias can propagate into subsequent iterations, overall convergence rate can degrade. Hence, in DPT-BO, we use PI, EI, and UCB in a sequential manner to benefit from each strategy without running into the biasing problem. The overall flowchart summarizing the DPT-BO method is given in Fig. 4.

TABLE I
PERFORMANCE OF DPT-BO ON TEST FUNCTIONS

	PSO		ADD-MES-G		DPT-BO	
	r_t	AUC (Normalized)	r_t	AUC (Normalized)	r_t	AUC (Normalized)
6D Hartmann	0.730	5.11	0.001	0.62	0.002	1.00
10D Schwefel	291.4	4.32	100.7	2.61	4.768	1.00
15D Levy	95.74	7.57	37.07	3.10	4.514	1.00
25D Qing	350.0	2.88	221.3	1.65	33.09	1.00

D. Experiments on Test Functions

In order to test the performance of DPT-BO, we have considered four test functions with different dimensionalities and response surface types that contain many local optima, which are common benchmarking problems in black-box optimization [46]. Performance evaluation criteria in these experiments are: 1) simple regret (r_t), the distance between actual global optima and the one found by the optimization algorithm and 2) area under curve (AUC), a measure of how fast an algorithm converges. AUC and r_t can be written as

$$r_t = |f^* - \tilde{f}_t^*|, \quad \text{AUC} = \frac{1}{T} \sum_{i=1}^T r_i \quad (24)$$

where f^* is the known global optimum of $f(x)$ and $\tilde{f}_t^*$ is the best value found by the optimization algorithm at t th iteration.

We compare DPT-BO with the widely used particle swarm optimization (PSO) and a recent HDBO method, maximum entropy search with additive GP (ADD-MES-G) [47], which uses the kernel in (13) along with a structural kernel learning strategy [19].

Results of these experiments are provided in Fig. 5 and in Table I. Except for the 6-D Hartmann function, the proposed DPT-BO algorithm results in a significantly better regret value than both ADD-MES-G and PSO. In addition, normalized AUC values show that, on average, DPT-BO converges $1.99\times$ and $4.97\times$ faster than ADD-MES-G and PSO, respectively. Table II further provides CPU times required to select the next sampling point for each technique. As PSO does not use a surrogate model, its run time is considerably faster compared to both ADD-MES-G and DPT-BO. Although there is no significant difference between run times of ADD-MES-G and DPT-BO, the sampling strategy of ADD-MES-G for structural kernel learning and estimating the acquisition function makes it slower as compared to DPT-BO.

It should be noted that the choice of hyperparameters can significantly affect the performance of BO-based algorithms, including DPT-BO. Here, problem specific knowledge can be leveraged to adjust these parameters. For instance, η in (11) can be decreased to give more importance to exploration than exploitation. Following the rule of thumb values [14], for these experiments, the hyperparameters of DPT-BO are chosen as

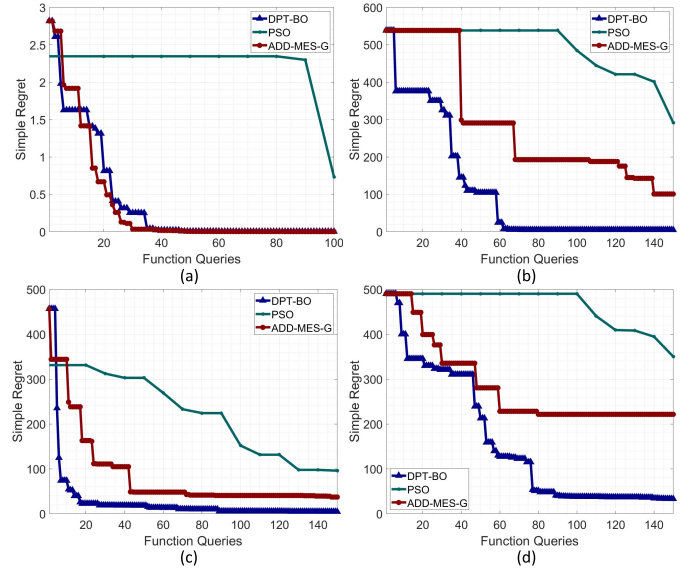


Fig. 5. Performance comparison of the proposed algorithm, DPT-BO, on optimization test functions. (a) 6-D Hartmann. (b) 10-D Schwefel. (c) 15-D Levy. (d) 25-D Qing. Functions are available in [46].

TABLE II
RUN TIMES FOR SELECTING THE NEXT SAMPLING POINT

	PSO	ADD-MES-G	DPT-BO
6D Hartmann	0.028s	0.324s	0.379s
10D Schwefel	0.032s	0.832s	1.205s
15D Levy	0.034s	1.701s	1.348s
25D Qing	0.034s	4.561s	2.501s

$\eta = 0.05$ in (11); $\zeta = 0.01$ and $\zeta = 0.1$ in (9) and (10) respectively; $n = 1, 2$, D in (15) and $d = 3, 5, 5, 5$ and $M = 2, 2, 3, 5$ in (16) for 6-D Hartmann, 10-D Schwefel, 15-D Levy, and 25-D Qing functions, respectively.

In order to quantify the effect of number of groups and maximum number of parameters in a group [d and M in (16)] on the performance of DPT-BO, an additional experiment is performed on the 15-D Levy function. Here, d is varied from 1 to 8 with the corresponding M values varying between 15 and 2. Results of this experiment are given in Fig. 6. The convergence is observed to be the worst when $d = 1$ and $d = 8$ due to over exploration for the former case and over exploitation for the latter as explained in Section III-C. The convergence for the $d = 5$ case is observed to be better compared to the case with $d = 3$. The reason is when d increases, the sensitivity of parameters becomes less important in the horizontal expansion strategy. This provides an advantage since the prioritization is less affected by the bias introduced in the early iterations of the optimization process, when the number of observations is not sufficient enough to accurately group the parameters. It should be noted that since the acquisition functions used already address the exploration and exploitation tradeoff, all choices of d and M have a certain level of balance. The parameters d and M thus cannot provide pure exploitation or pure exploration. Following this analysis, we choose $d = 3, 5, 6$ and $M = 3, 3, 5$ for the design applications in Sections IV–VI, respectively.

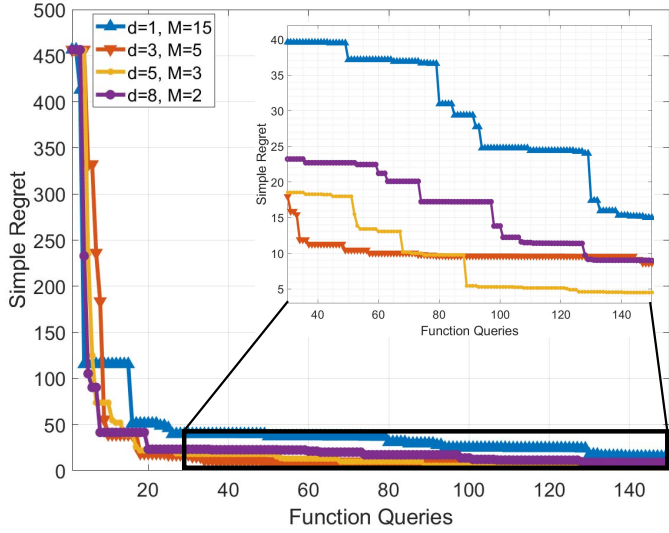


Fig. 6. Effect of number of groups $[d]$ and M in (16) on performance of DPT-BO on minimizing 15-D Levy function.

IV. APPLICATION I: INTERCONNECTS IN HIGH-SPEED CHANNELS

The first application chosen to evaluate the proposed algorithm is the optimization of interconnects in high-speed chip-to-chip interconnect channels. Optimization of such high-speed channels is a nontrivial task given that the system has a nonlinear response to its control parameters such as the geometrical parameters of the interconnects and the material properties. Furthermore, characterizing electrical properties of such interconnects requires a full-wave EM simulation over a very large frequency bandwidth, i.e., DC to high GHz regimes, followed by a bit-by-bit time-domain simulation with a large pseudorandom bit sequence (PRBS) to characterize the eye diagram at very low bit error rate (BER) contours.

In order to eliminate frequency sweeps over a large bandwidth from the optimization loop, we use the Efficient Bayesian Framework (EBF) as described in [48] and first build a parametric surrogate model of the frequency response of the channel. In EBF, frequency is considered as an additional input parameter. Hence, the training data required to build the model can be acquired by performing simulations at single frequency points, thereby greatly reducing the CPU time required to build the surrogate model. This model is then used in the optimization loop, along with a time-domain simulation, to directly optimize the eye opening. Note that although the surrogate model replaces the EM solver to characterize the frequency response, the time-domain simulation is still CPU intensive. The high-speed channel considered in this section is three single-ended embedded microstrip lines on a silicon interposer referenced to a nonideal power/ground (P/G) grid as shown in Fig. 7. The input parameters comprising a 10-D sample space are given in Table III.

A. Channel Surrogate Model

In order to create the surrogate model of the microstrip channel, we use a GP with an SE kernel. Since the

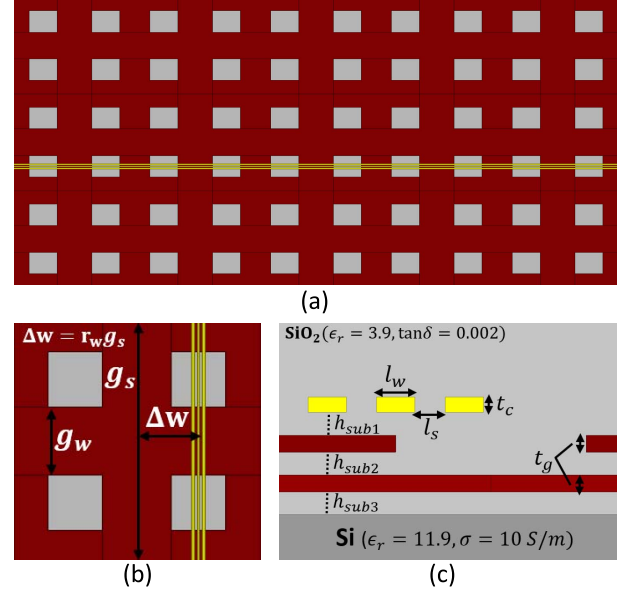


Fig. 7. Structure of the high-speed channel considered. (a) Top view of overall channel. (b) Unit cell. (c) Cross section.

TABLE III
CONTROL PARAMETERS OF THE HIGH-SPEED CHANNEL

Parameter	Unit	Min	Max
Line Width	l_w	μm	0.4 3
Line Thickness	t_c	μm	0.4 3
Line Spacing	l_s	μm	0.4 3
Substrate Height I	h_{sub1}	μm	1 5
Substrate Height II	h_{sub2}	μm	1 5
Substrate Height III	h_{sub3}	μm	1 5
P/G Grid Width	g_w	μm	10 40
P/G Grid Thickness	t_g	μm	0.4 3
Signal Misalignment Ratio	r_w		0 0.5
Frequency	f	GHz	0.05 20

transmission lines considered in Fig. 7(a) have discontinuities in its reference plane, the propagation constant varies along the direction of propagation. Hence, per unit length RLGC matrices cannot be defined. We therefore consider the unit cell as in Fig. 7(b) and model per unit cell (p.u.c) RLGC matrices of the microstrip channel.

First, 1000 samples based on uniform Latin hypercube sampling (LHS) are determined. Then, a full-wave EM solver, Ansys HFSS, is used to extract the p.u.c RLGC matrices. As we consider frequency as an input to the surrogate model, simulations are done at single frequency points rather than a sweep in the entire bandwidth. Here, each set of geometrical parameters in the training data is evaluated at different frequency points. Hence, the total amount of CPU time required to collect training data is greatly reduced. Note that as any other geometrical parameter given in Table III, frequency points follow a uniform marginal distribution as determined by LHS. The dimensionality of the output space defined by the full rank p.u.c RLGC matrices is 36. Since the microstrip channel is a reciprocal network, it is sufficient to consider the elements in the lower triangular and the diagonal of the RLGC matrices, effectively reducing the number of outputs to 24.

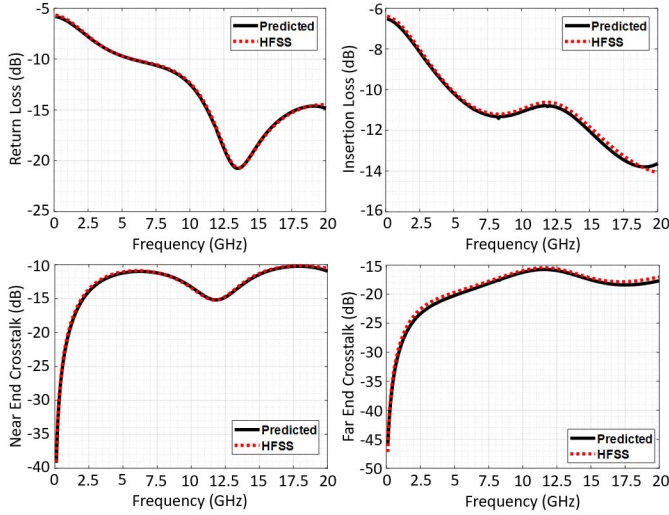


Fig. 8. Comparison of the GP model with full-wave simulations for a channel of length 10 mm.

TABLE IV
NMSE VALUES OF THE GP MODEL

Parameter	NMSE	Parameter	NMSE	Parameter	NMSE	Parameter	NMSE
R ₁₁	0.0003	L ₁₁	0.0212	G ₁₁	0.0011	C ₁₁	0.0041
R ₁₂	0.0291	L ₁₂	0.0009	G ₁₂	0.0013	C ₁₂	0.0079
R ₁₃	0.0004	L ₁₃	0.0075	G ₁₃	0.0009	C ₁₃	0.0023
R ₂₂	0.0184	L ₂₂	0.0003	G ₂₂	0.0014	C ₂₂	0.0057
R ₃₂	0.0291	L ₃₂	0.0012	G ₃₂	0.0096	C ₃₂	0.0030
R ₃₃	0.0003	L ₃₃	0.0301	G ₃₃	0.0163	C ₃₃	0.0049

Since the RLGC models are to be used in a time-domain simulation, predictions need to preserve physical consistency, i.e., causality and stability, of the interconnect model. For a lossy, linear time-invariant (LTI) system, passivity is a sufficient condition for causality and stability [49]. For an RLGC network to be passive, each element in these matrices has to have a certain property. Diagonal terms of C and G matrices have to be nonnegative and off-diagonal terms must be nonpositive. Furthermore, each element of R and L matrices has to be nonnegative. These properties can be preserved by using a *log* transformation on the training data as a preprocessing step. This can then be reversed in the prediction stage by exponentiation of the predicted values, thereby ensuring every element in predicted RLGC matrices has the proper sign that ensures passivity and physical consistency.

After the training data are collected and preprocessed using log transformation, 750 out of 1000 samples are standardized to have zero mean and unit standard deviation and used to perform the training of the GP model to maximize log marginal likelihood as explained in Section II. Since GP models with SE kernel can only model a single output at a time, 24 independent GP models have been trained to predict p.u.c RLGC matrices.

The remaining 250 samples are then used to assess the quality of the GP model using the normalized mean-squared

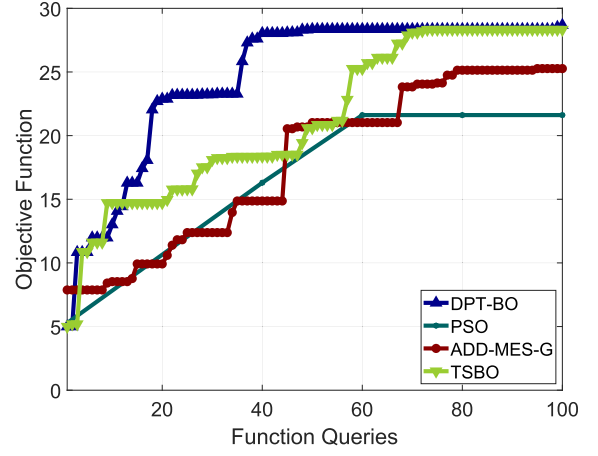


Fig. 9. Performance of the proposed algorithm for maximizing the objective function in (26).

TABLE V
OPTIMIZATION RESULTS FOR HIGH-SPEED CHANNEL

	PSO	TSBO	ADD-MES-G	DPT-BO
Eye Width (ps)	50.31	54.15	52.18	53.13
Eye Height (V)	0.46	0.52	0.48	0.54
Pk-Pk Jitter (ps)	10.3	8.04	8.26	8.12
AUC (Normalized)	1.48	1.19	1.41	1.00

error (NMSE) criteria, given as

$$\text{NMSE} = \frac{\sum_{i=1}^N (\hat{y}(x) - y(x))^2}{\sum_{i=1}^N \left(y(x) - \frac{1}{N} \sum_{i=1}^N y(x) \right)^2} \quad (25)$$

where $N = 250$; $y(x)$ is the validation data and $\hat{y}(x)$ represent predictions done by the GP model at validation samples. As can be seen from Table IV, the NMSE for p.u.c RLGC matrices is kept less than 3.01%, showing the high quality of the predictive model. A further test for the model accuracy is performed by converting predicted RLGC matrices to S-parameters of the unit cell. This is then cascaded to itself to form the frequency response of a channel with a length of 10 mm. For a random parameter assignment, the comparison for insertion loss (IL), return loss (RL), near-end (NEXT), and far-end (FEXT) crosstalk over the entire bandwidth can be seen in Fig. 8.

B. Optimization of Interconnects

The objective function for the direct eye optimization is defined as

$$f(x) = W_E H_E \quad (26)$$

where W_E and H_E denote the width and the height of the eye diagram at a BER contour of 10^{-12} . The optimization loop starts by determining the geometrical parameters of the microstrip channel by the optimization algorithm while fixing the P/G grid spacing, i.e., $g_s = 50 \mu\text{m}$ in Fig. 7(b) to fix the length of the unit cell. The p.u.c RLGC matrices over the entire

frequency bandwidth are then determined by the GP model, converted to S-parameters and then cascaded to each other to form the interconnect channel with a length of 10 mm. The channel S-parameters are then used by a commercial circuit solver, Keysight ADS, to perform bit-by-bit simulation and generate the eye diagram at a data rate of 16 Gbps. The output impedance of the driver is fixed to 50Ω and the load at the receiver is fixed as a 50Ω resistor with a shunt capacitor of 1 pF to represent the pad parasitics. The eye width and height generated by ADS is then combined in (26) and fed back into the optimization algorithm to proceed into next iteration.

The optimization results are summarized in Table V and in Fig. 9. Optimization using DPT-BO resulted in an eye width and height of 53.13 ps and 0.54 V along with a peak-to-peak jitter of 8.12 ps, corresponding to a improvement of 1.8%, 11.1%, 1.7%, and 5.3%, 14.8% and 21.2% as compared to ADD-MES-G and PSO, respectively. In addition, the normalized AUC values show that DPT-BO converged $1.41\times$ and $1.48\times$ faster as compared to ADD-MES-G and PSO, respectively. For the sake of completeness, the BO approach that uses the partitioning tree with only vertical expansion, Two-Stage Bayesian Optimization (TSBO) [22], is also applied to maximize the eye opening. As shown in Table V, optimization using TSBO and DPT-BO has resulted in similar eye characteristics. However, DPT-BO converged $1.19\times$ faster than TSBO. As explained in previous sections, only vertical expansion lacks proper exploitation of the sample space, hence converges slowly as the problem dimensionality increases. This limits the use of TSBO in higher dimensional design problems considered in subsequent sections.

V. APPLICATION II: SUBSTRATE INTEGRATED WAVEGUIDE WITH AIR CAVITY

The second application chosen in this work is the optimization of a SIW with the goal of minimizing the losses over D-band. SIW technology is a very promising alternative to conventional microstrip and coplanar waveguide transmission lines as the operating frequency increases into the terahertz region. The limitation, however, is the increasing dielectric loss that tends to dominate the total transmission losses. To address this, SIWs designed on a multilayer PCB with an air cavity in the middle layer have attracted attention [50]. Optimization of such structures to minimize total transmission losses can be a very challenging problem given the nonlinearity of the parameters.

In this section, we consider the SIW structure with air cavity inside a liquid crystal polymer (LCP) substrate ($\epsilon_r = 3.1, \tan \delta = 0.005$ as shown in [51]) in Fig. 10 and use DPT-BO to automatically find the optimal shape of the air cavity along with the geometrical parameters of SIW and microstrip to SIW transition. The input parameters comprise a 14-D problem as shown in Table VI. Note that the length of the waveguide section is fixed to 2 mm. The air cavity is parameterized as two elliptical regions connected by a rectangular region as shown in Fig. 10. We consider two cases, namely,) no air cavity and air cavity. We compare the results with an all-LCP structure that was designed through hand tuning and fabricated [51].

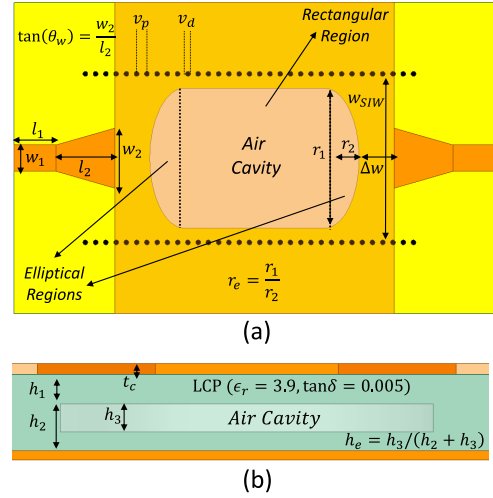


Fig. 10. Structure of the SIW with elliptical air cavity. (a) Top view. (b) Cross section.

TABLE VI
CONTROL PARAMETERS OF THE SIW STRUCTURE

Parameter		Unit	Min	Max
SIW Width	w_{SIW}	mm	0.5	2.5
Microstrip Width	w_1	mm	0.1	0.3
Microstrip Length	l_1	mm	0.2	1
Taper Angle	θ_w	deg	60	90
Taper Length	l_2	mm	0.2	1
Via Diameter	v_d	μm	20	60
Via Pitch	v_p	μm	50	150
Copper Thickness	t_c	μm	10	30
Substrate Height I	h_1	μm	10	30
Substrate Height II	h_2	μm	10	90
Air Cavity Major Radius	r_1	mm	0.1	10
Air Cavity Aspect Ratio	r_e	μm	0.1	10
Air Cavity Offset	Δw	mm	0.1	0.9
Air Cavity Ratio	h_e		0	1

A. Optimization Setup

The objective function to be maximized is chosen as the minimum magnitude of the transmission coefficient over D-band, given as

$$f(x) = \min(|S_{21}(f)|), \quad \text{where } f \in [110, 170] \text{ GHz.} \quad (27)$$

We rely on EM simulation to account for the arbitrarily shaped air cavity and microstrip to SIW transition while capturing the effect of variations in all geometrical parameters. We use Ansys HFSS to extract the S-parameters of the SIW to model the transmission losses in this paper.

B. Results

The optimization results are summarized in Table VII and Fig. 11. Optimization using DPT-BO resulted in minimum $|S_{21}|$ value of -0.763 dB over D-band as compared to -1.082 and -1.163 dB using ADD-MES-G and PSO, corresponding to a 29.5% and 34.4% improvement in transmission losses, respectively. Though all algorithms started from the same initial point, normalized AUC values show that DPT-BO converged $1.35\times$ and $1.42\times$ faster as compared to ADD-MES-G and PSO, respectively.

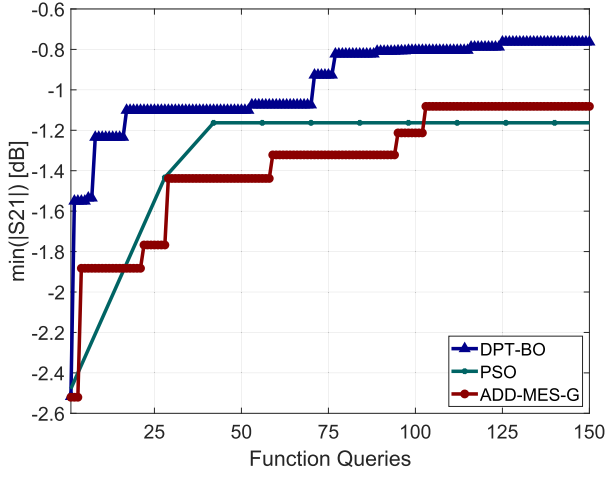


Fig. 11. Performance of the proposed algorithm on loss minimization of SIW.

TABLE VII
OPTIMIZATION RESULTS FOR SIW

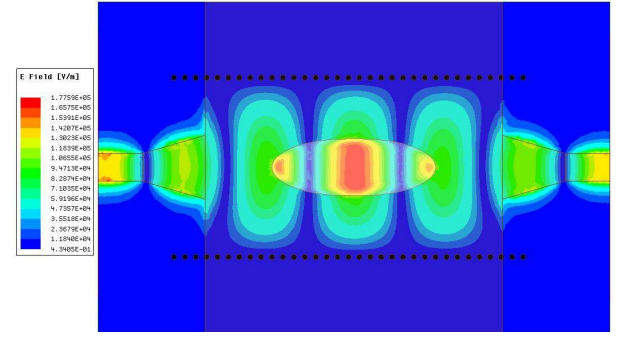
	PSO	ADD-MES-G	DPT-BO
Min. $S_{21}(f)$ (dB)	-1.163	-1.082	-0.763
AUC (Normalized)	1.35	1.42	1.00

To quantify the advantage of using an air cavity inside the LCP substrate, we use DPT-BO to optimize the SIW design without an air cavity and compare its performance with the design with the optimal air cavity as well as the performance of the hand-tuned SIW with LCP in [51]. The minimum IL for the optimal LCP SIW design is found to be -1.159 dB at 170 GHz and -1.911 dB for the hand-tuned design, corresponding to 34.2% and 60.1% worse performance compared to DPT-BO optimized design with the air cavity. This also shows that DPT-BO optimized LCP SIW has 39.4% better performance than the hand-tuned LCP SIW using the same structure.

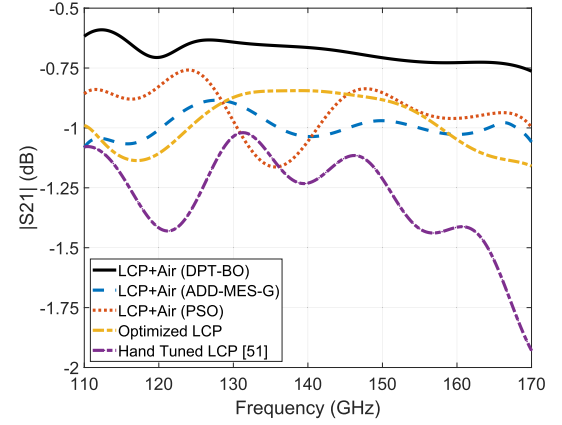
Fig. 12 shows that the optimal LCP-only design has comparable performance with the designs optimized using both ADD-MES-G and PSO. This shows that both of these algorithms were stuck in a local maxima when maximizing the objective function in (27), whereas DPT-BO was able to get out of the local maxima and provide a better final design. Furthermore, the E -field distribution and $|S_{11}|$ in Fig. 12(a) and (c) show that the reflections caused by the transition from LCP to air cavity are minimized to minimize the transmission loss. Cutoff frequencies for TE_{10} and TE_{20} modes of the design optimized by DPT-BO are simulated as $f_{c10} = 81.3$ GHz and $f_{c20} = 161.8$ GHz, showing losses due to dispersion and mode conversion are also minimized.

VI. APPLICATION III: WIRELESS POWER TRANSFER

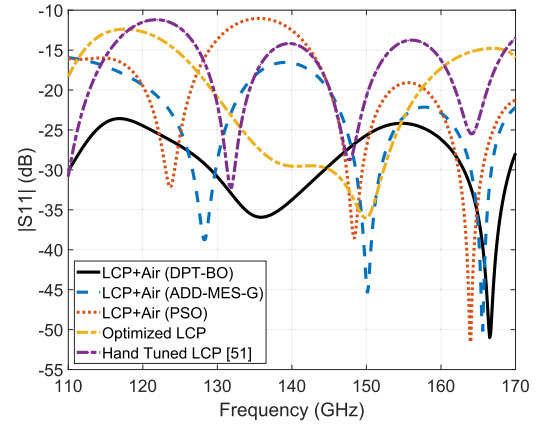
The final application considered to evaluate the performance of the proposed DPT-BO algorithm is optimization of an inductive coupling-based WPT system operating at 1 GHz. The architecture of the WPT system is given in Fig. 13,



(a)



(b)



(c)

Fig. 12. Performance of the final SIW design. (a) E -field distribution at 170 GHz showing the optimal air cavity. (b) $|S_{21}|$ comparison for different algorithms. (c) $|S_{11}|$ comparison for different algorithms.

which represents an integrated board solution, consisting of embedded rectangular RF coils as shown in Fig. 14 connected to TX & RX matching networks, a full-bridge diode rectifier and a buck converter (BC) for DC regulation [52]. The targeted application is efficient power delivery to IoT devices requiring input power in the range of 1–100 mW.

Two major design challenges in such power delivery architectures are to maximize conversion efficiency to reduce battery wastage and minimizing the area of the system to be mounted on the IoT device. In this section, we address the design complexity by formulating the design procedure as a system-level multiobjective optimization problem with

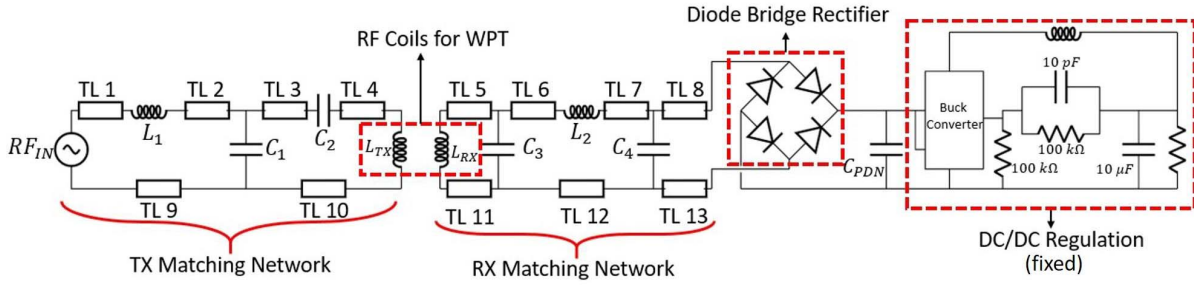


Fig. 13. Schematic of the WPT-based power delivery architecture [34].

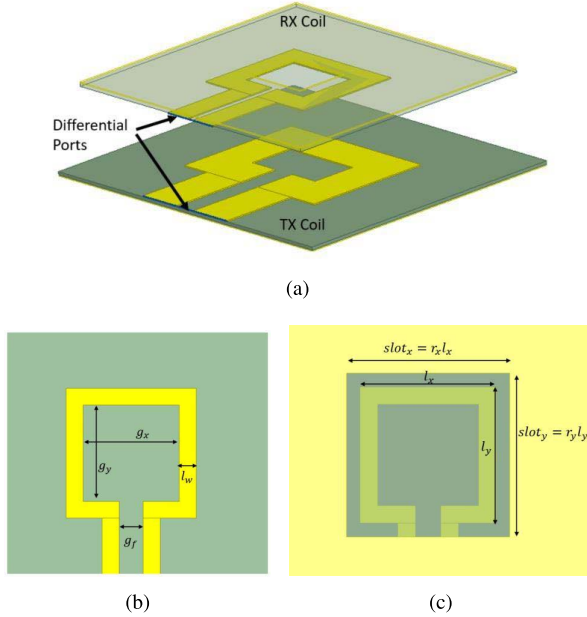


Fig. 14. Geometry of the embedded RF coils defined by the control parameters [34]. (a) WPT structure. (b) Top view. (c) Bottom view.

the goal of maximizing the RF-DC conversion efficiency and minimizing the area of the RX coil. Previous work demonstrated a stage-by-stage optimization procedure for the same architecture considered in this paper [34]. However, as common in RF systems, individual blocks in the WPT architecture are coupled to each other and have contradicting design tradeoffs, which raises the necessity to perform system-level optimization. This translates the optimization problem into a very high-dimensional one along with a highly nonlinear response surface with many local optima. The input parameters comprising a 32-D sample space are given in Table VIII.

A. Optimization Setup

The total system efficiency can be calculated by cascading the rectifier and DC-DC regulation efficiencies. Maximizing rectifier efficiency results in increased voltage on the load of the rectifier, i.e., input impedance of the BC. However, the BC requires lower input voltage to minimize the switching losses. Since the design tradeoff between RF-DC conversion and DC-DC conversion stages is limited with this phenomena, we fix the DC regulation stage and focus on maximization of rectifier efficiency while minimizing the input voltage of the

TABLE VIII
CONTROL PARAMETERS OF WPT ARCHITECTURE

Parameter		Unit	Min	Max
Inner Height of TX coil	$g_{y,TX}$	mm	1	5
Inner Height of RX coil	$g_{y,RX}$	mm	1	5
Inner Width of TX coil	$g_{x,TX}$	mm	1	5
Inner Width of RX coil	$g_{x,RX}$	mm	1	5
Line Width of TX coil	$l_{w,TX}$	mm	0.5	3
Line Width of RX coil	$l_{w,RX}$	mm	0.5	3
Feeding Gap for TX coil	$g_{f,TX}$	mm	0.5	3
Feeding Gap for RX coil	$g_{f,RX}$	mm	0.5	3
TX Vertical GND Cut-out Ratio	$slot_{y,TX}$		0.8	1.2
RX Vertical GND Cut-out Ratio	$slot_{y,RX}$		0.8	1.2
TX Horizontal GND Cut-out Ratio	$slot_{x,TX}$		0.8	1.2
RX Horizontal GND Cut-out Ratio	$slot_{x,RX}$		0.8	1.2
Capacitor I	C_1	pF	0.1	10
Capacitor II	C_2	pF	0.1	10
Capacitor III	C_3	pF	0.1	10
Capacitor IV	C_4	pF	0.1	10
Inductor I	L_1	nH	0.1	10
Inductor II	L_2	nH	0.1	10
Input Power	$P_{RF,IN}$	dBm	5	15
Widths of all TLINs (13 separate parameters)	$w_{TL1,...,TL13}$	mil	15	45

BC. To minimize the overall size of the WPT architecture, lengths of all transmission lines are fixed to 0.5 mm based on process capabilities and the area of RX coil is minimized. Hence, the system-level optimization is formulated as maximizing the weighted sum of these objectives, given as

$$f(x) = \sum_{i=1}^3 w_i y_i \quad (28)$$

where y_1 , y_2 , and y_3 are rectifier efficiency, minimum input voltage of the BC, and area of RX coil; $w_1 = 7$, $w_2 = -3.5$, and $w_3 = -3$ are the corresponding weights, chosen to prioritize efficiency over area. The BC considered in this section requires a minimum of 2.5 V as input voltage [53]. Hence, y_2 is written as $|V_{BC, IN} - 2.5 \text{ V}|$ to set the target BC input voltage to 2.5 V.

The simulation framework used starts with a full-wave EM simulation (Ansys HFSS) to extract the S-parameters of the RF coils that capture the impedance profile and the coupling between them. The S-parameters are then fed into Keysight ADS to perform harmonic balance simulation to calculate the rectifier efficiency. The transmission distance between RF coils

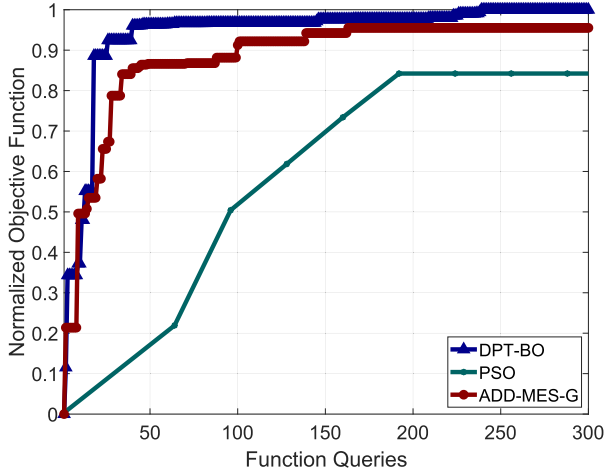


Fig. 15. Performance comparison of the proposed algorithm on maximizing objective function in (28).

TABLE IX
OPTIMIZATION RESULTS FOR WPT SYSTEM

	PSO	ADD-MES-G	DPT-BO
RX Coil Area (mm²)	7.48	19.26	11.04
Rectifier Efficiency	53.25	65.72	66.96
Input Voltage of BC (V)	3.24	2.61	2.72
System Efficiency	45.83	58.86	59.57
AUC (Normalized)	1.50	1.07	1.00

is fixed to 1 mm and the load of the BC, i.e., IoT device, is selected to operate at 1.2 V with 7.2 mW input power. The simulation-measurement correlation for our framework can be found in [34].

B. Results

The optimization results are summarized in Fig. 15 and Table IX. Optimization using DPT-BO resulted in 66.96% rectification efficiency, along with RX coil area of 11.04 mm² and BC input voltage of 2.72 V, corresponding to 58.86% total system efficiency. Although the electrical performance of the design found by ADD-MES-G is very similar to the one with DPT-BO, it provided an RX coil area of 19.26 mm², corresponding to 42.7% larger area as compared to DPT-BO. PSO provided an RX coil area of 7.48 mm², however, it performed significantly worse in terms of system efficiency with 13.03% and 13.74% reduced efficiency compared to ADD-MES-G and DPT-BO, respectively. The normalized AUC values show that DPT-BO converged 1.50 $\times$ and 1.07 $\times$ faster than PSO and ADD-MES-G, respectively.

In addition, Fig. 16 shows the breakdown of the objective function in (28) to its three components of rectifier efficiency, BC input voltage and RX coil area. This breakdown shows the design tradeoffs of each method. Both ADD-MES-G and DPT-BO have tradedoff rectifier efficiency to reduce BC input voltage and minimize RX coil area, whereas PSO overreduced

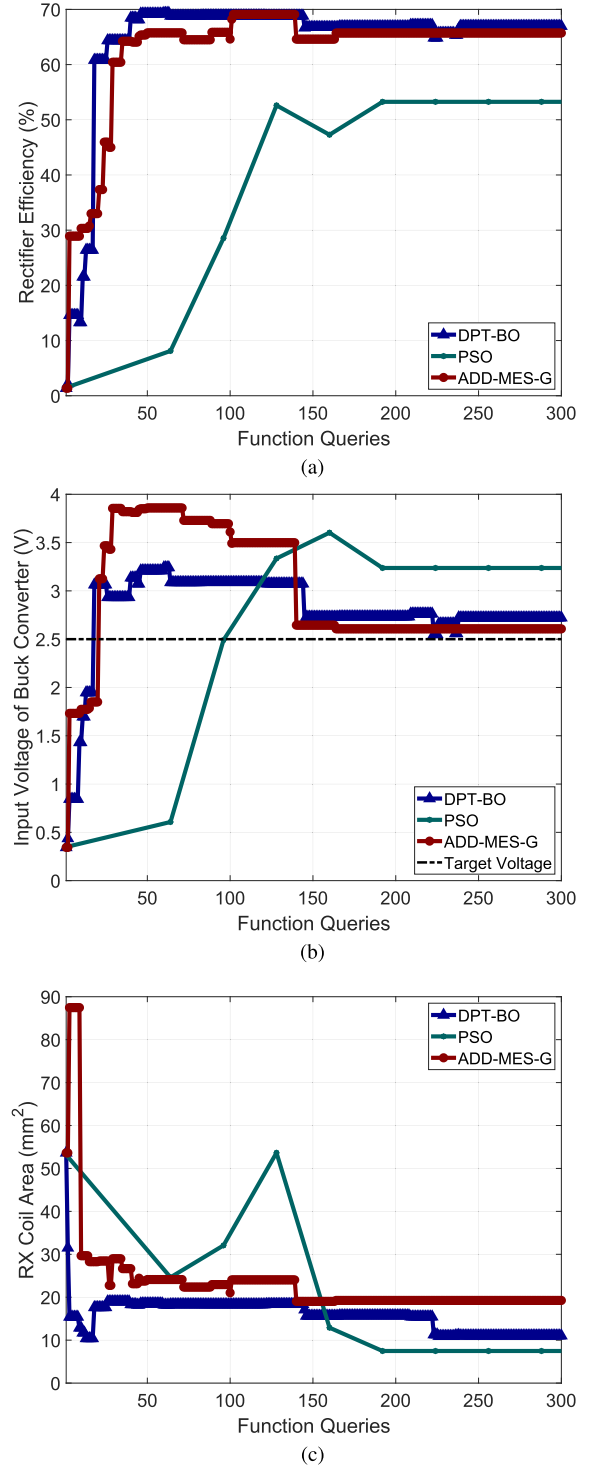


Fig. 16. Convergence comparison for the three components of the objective function in (28). (a) Rectifier efficiency. (b) Input voltage of BC. (c) RX coil area.

the rectifier efficiency to minimize the area and got stuck in a local maxima.

VII. CONCLUSION

In this paper, we have proposed a new BO-based global optimization method, Bayesian Optimization with Deep Partitioning Tree, that is designed for high-dimensional problems. This is

accomplished by utilizing an ADD-GP as the surrogate model. Unlike previous methods in the literature that uses ADD-GP with the disjoint group assumption, we have considered a fully additive decomposition to make DPT-BO better suited for high-frequency design optimization problems. In order to rapidly cover the high-dimensional sample space and eliminate auxiliary optimization, we have proposed and used a Deep Partitioning Tree. The proposed DPT-BO algorithm has been applied to a set of optimization test functions with different dimensionalities and response surface types. The proposed method showed better performance results with faster convergence.

DPT-BO has also been applied to three applications arising in high-frequency electronic design, namely: 1) 9-D optimization of interconnects in high-speed channels to maximize eye diagram; 2) 14-D optimization of a SIW structure with air cavity inside an LCP substrate operating in D-band; and 3) 32-D multiobjective optimization of a wireless power transfer based power delivery system to maximize the system efficiency. The results presented in this paper show the applicability of DPT-BO to perform global optimization on a variety of high-dimensional designs that operate in various frequency bands and therefore provide for a generic solution.

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